Global Electronic Retailer: SQL Sales Analytical Insights

Business Summary

The business has been operating since **January 1, 2016,** and the dataset covers sales up to **February 20, 2021**.

The company sells **2,517 electronic product models** across **32 product types (subcategories)**, grouped into **8 major product categories**. Sales occur through both **online channels** and **physical stores**, spanning **8 different countries and Ecom**.

KPIs

Metric	Value
Total Revenue	55.8 M
Total Orders	26.3 K
Total Profit	32.7 M
Total Quantity Sold	197757
Average Order Value	2118
Total customers	15266
Active Customers (who placed orders)	11887
Average Unit Per Order	7.5

Over the **5-year** operating period (Jan 2016 – Feb 2021), the business generated **$55.7M** in **total revenue** and **$32.6M in profit**, reflecting a 58.6% overall margin.
(a figure that should be read with the caveat that this dataset's product pricing is synthetically generated at a consistent cost-to-price and is higher than typical real-world electronics retail margins.)

Of the **15,266** registered **customers, 11,887 (77.9%)** have placed at least one order, meaning nearly a quarter of the customer base (22.1%) has never converted to a purchase — a potential area for targeted re-engagement or onboarding follow-up. The **average order** includes **7.5 units** at a **value of $2,117.89,** across **26,326 total orders** and **197,757 units sold**.

Sales Trend Over Time

Year	Orders	Units Sold	Avg Order Value	Revenue	Profit	Profit %	Period
2021	498	3,797	2,086.92	**1.04M**	**0.61M**	58.55%	YTD Feb
2020	4,635	34,463	2,005.31	**9.29M**	**5.45M**	58.61%	Full Year
2019	9,083	68,440	2,010.83	**18.26M**	**10.70M**	58.57%	Full Year
2018	5,965	44,498	2,144.00	**12.79M**	**7.46M**	58.37%	Full Year
2017	3,280	24,798	2,262.63	**7.42M**	**4.34M**	58.44%	Full Year
2016	2,865	21,761	2,424.71	**6.95M**	**4.11M**	59.12%	Full Year

Year	Orders	Orders YoY %	Revenue (USD)	Revenue YoY %	Profit (USD)	Profit YoY %
2020	4,635	**−48.97%**	**9.29M**	**−49.11%**	**5.45M**	**−49.08%**
2019	9,083	**52.27%**	**18.26M**	**42.81%**	**10.70M**	**43.31%**
2018	5,965	**81.86%**	**12.79M**	**72.32%**	**7.46M**	**72.12%**
2017	3,280	**14.49%**	**7.42M**	**6.83%**	**4.34M**	**5.60%**
2016	2,865	—	**6.95M**	—	**4.11M**	—

The business grew steadily from 2016 through 2019, with orders climbing from **2,865 to 9,083** (a cumulative increase of over 200%) and revenue nearly tripling from **$6.9M to $18.3M**. Growth was strongest in 2018 **(+82% orders, +72% revenue**), before decelerating somewhat in 2019 **(+52% orders, +43% revenue**) — still strong growth, just a slower rate of increase. 2020 marks a sharp reversal, with orders, revenue, and profit all declining by roughly 49% year-over-year, reflecting external market disruptions - a significant shift worth flagging as a key finding, though the dataset doesn't provide contextual information.

(Note: Data for 2021 is excluded from this year-over-year comparison, as the dataset ends February 20, 2021 — a partial year would distort the comparison against full prior years.)

Geographic & Channel Analysis

Country	Revenue	% of Total
United States	23.8M	43%
Online	11.4M	20%
United Kingdom	5.7M	10%
Germany	4.2M	8%
Canada	3.6M	6%
Australia	2.1M	4%
Italy	2.1M	4%
Netherlands	1.6M	3%
France	1.2M	2%

Revenue is heavily concentrated in the United States, generating **43%** of all global sales. While this represents massive market dominance, it also introduces a high level of operational risk.

Treating the **Online** e-commerce store as a standalone entity reveals it is the second largest "market," driving an incredible **20%** of total revenue. Digital sales outperform every single international brick-and-mortar territory outside of the US.

The low performance in France (2%) and the Netherlands (3%) signals either an under-penetrated market or poor local execution. A market assessment should be conducted to evaluate whether to increase local localized marketing or transition lagging physical stores in these regions entirely into the high-performing Online e-commerce infrastructure.

Top & Bottom Performers

Top 5 Products by Revenue

Product	Revenue
WWI Desktop PC2.33 X2330 Black	$505K
Adventure Works Desktop PC2.33 XD233 Silver	$466K
Adventure Works Desktop PC2.33 XD233 Brown	$464K
Adventure Works Desktop PC2.33 XD233 Black	$448K
Adventure Works Desktop PC2.33 XD233 White	$437K

High-end desktop computing systems act as the primary cash-flow anchors for the business. These products should remain a priority for inventory planning, marketing campaigns, and forecasting

Bottom 5 Products by Revenue

Product	Revenue
Contoso Home Theatre System 5.1 Channel M1530 White	$0
Contoso Washer & Dryer 21in E210 Green	$0
Adventure Works Floor Lamp X1150 Black	$0
Adventure Works Floor Lamp M2150 Black	$0
Adventure Works Chandelier M6150 Black	$0

These products represent dead weight in the catalogue. In a live retail environment, carrying zero-velocity stock creates warehouse overhead and ties up capital. This finding dictates an immediate operational recommendation to halt manufacturing or prune these SKUs from the active global catalogue.

Top 5 Customers by Revenue

Customer	Revenue
Matthew Flemming	$61.9K
Karen Jones	$43.5K
Zrina Topic	$42.8K
Stefanie Hartmann	$41.5K
Stephan Rothstein	$40.6K

Led by Matthew Flemming with over $61K in personal lifetime purchases, these top 5 accounts represent a critical concentration of value. This specific output list can be directly routed to marketing engines to seed high-value retention tracks, exclusive early product access, or premium customer support layers to shield these vital accounts from competitor churn.

Customer Segmentation

Customer Segmentation	Total Customers
New	6450
VIP	2800
Regular	2637

High-Volume Acquisition Runway (New Customers: These customers have been with the business for less than 12 months**) -** The largest segment of active buyers consists of **New Customers** (representing 54.3% of the active database). This massive influx proves that the business has had highly effective front-end marketing and customer acquisition strategies over its operating lifecycle.

The Core Revenue Engine (VIP Customers: VIP customers have at least 12 months of history and have spent more than $5,000.**) -** A massive **2,800 customers** have scaled into the elite **VIP** tier. In real-world retail operations, having nearly a quarter (23.6%) of your active customer base qualify as high-value, high-spend VIPs is exceptionally strong. These users should be funnelled into exclusive loyalty programs, offered premium priority support, and targeted with early-access hardware campaigns to shield them from competitor churn—especially during macro-market volume contractions.

The Stable Baseline (Regular Customers: These customers have at least 12 months of history but have spent $5,000 or less.) **-** Sitting right between your high-value VIPs and raw newcomers are **2,637 Regular Customers** (22.2%). These are your consistent, predictable buyers who know the brand and return periodically.

Category Contribution

Category	Total Sales	% of Total
Computers	19.3M	34.60%
Home Appliances	10.8M	19.40%
Cameras & Camcorders	6.5M	11.70%
Cell Phones	6.2M	11.10%
TV & Video	5.9M	10.60%
Audio	3.2M	5.70%
Music, Movies & Audio Books	3.1M	5.60%
Games & Toys	0.7M	1.30%

Computers is the dominant revenue category, contributing 34.6% of total sales, more than the next two categories combined. This aligns perfectly with our previous discovery that premium desktop computing hardware represents the primary top-line cash anchor for the business.
Home Appliances represents nearly a fifth of total turnover. Together with Computers, these two core categories control **over 54%** of everything the company sells.
Cameras & Camcorders **(11.70%)**, Cell Phones **(11.10%)**, and TV & Video **(10.60%)** form a highly balanced mid-tier cluster, bringing in a tight range of $5.9M to $6.5M each. Cross-selling campaigns should target these mid-tier buyers with bundles.
At just **$0.7M**, Games & Toys is completely negligible to the bottom line. It captures almost no meaningful market share for the enterprise.

Data Quality Notes

Several data cleaning steps were performed during the SQL data warehouse build to resolve source file inconsistencies, including mixed character encodings across files, malformed CSV fields, and inconsistent currency formatting. Full technical details of these fixes are documented in the accompanying data warehouse README.
One finding directly affects how figures in this report should be interpreted: the 58-59% profit margin reflects the dataset's synthetic construction rather than typical real-world retail economics. Product unit price values are generated as a consistent multiple (1.9-2.0x) of unit cost across the entire catalogue, producing a uniformly high margin regardless of category. Real-world electronics retail margins are typically much thinner (5-35%, depending on category). This figure should be read as a characteristic of the dataset, not a reflection of actual market conditions.

Key Takeaways

Strong historical growth, recent decline. Revenue grew from $6.9M (2016) to $18.3M (2019), before falling 49% in 2020 — a decline consistent with widely-reported global disruptions to retail that year, though not directly confirmed by this dataset. Early 2021 data show signs of stabilization, with February 2021 revenue and profit both slightly ahead of February 2020.
Revenue is concentrated, not diversified. The United States alone accounts for 43% of total revenue, with the Online channel contributing a further 20% — together, over 60% of the business rests on two revenue sources. Similarly, Computers alone drives 34.6% of category revenue, with the top two categories accounting for over half of all sales.
Customer value is healthily distributed, not top-heavy. Nearly a quarter (23.6%) of active customers qualify as VIP-tier (12+ months history, $5,000+ spend), suggesting a genuinely loyal core customer base rather than reliance on a handful of outlier accounts.

Conclusion

This analysis demonstrates a business with a strong historical growth trajectory and a loyal, meaningfully-sized high-value customer segment, but with notable concentration risk across both geography (US/Online dominance) and product category (Computers dominance). The 2020 downturn and early-2021 stabilization warrant continued monitoring, and the zero-sales product tail and non-converting customer segment represent immediate, actionable opportunities for operational and marketing follow-up.
This report was produced entirely through SQL Server queries against a custom-built Bronze-Silver-Gold data warehouse, demonstrating end-to-end capability from raw data ingestion through business insight generation.

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